SQL Data Analysis

- Student: Shahd Reda
- Dataset: Sales Data

Query 1: Filter Records Using WHERE

SQL Query:

SELECT *

FROM Sales

WHERE UnitPrice > 500

LIMIT 10;

Output:

	OrderID	Data	CustomerID	Product	Quantity	UnitPrice	ShippingAddress	PaymentMethod	OrderStatus	TrackingNumber	ItemsInCart	CouponCode	ReferralSource	TotalPrice
1	ORD200000	1/2/1900	C72649	Monitor	5	570.62	928 Main St	Debit Card	Shipped	TRK37947903	7	SAVE10	Instagram	2853.1
2	ORD200002	1/6/2023	C81728	Tablet	5	550.68	512 Main St	Credit Card	Cancelled	TRK42903982	8	FREESHIP	Email	2753.4
3	ORD200004	1/8/2023	C81840	Printer	4	626.01	668 Main St	Online	Delivered	TRK29241424	8	SAVE10	Email	2504.04
4	ORD200006	1/10/2023	C83492	Laptop	1	664.42	986 Main St	Gift Card	Returned	TRK96417362	6	SAVE10	Facebook	664.42
5	ORD200009	1/13/2023	C31946	Desk	4	509.38	102 Main St	Credit Card	Shipped	TRK33478363	6	WINTER15	Google	2037.52
6	ORD200010	1/14/2023	C43443	Tablet	5	625.97	333 Main St	Credit Card	Returned	TRK98859248	9	SAVE10	Instagram	3129.85
7	ORD200016	1/20/2023	C28976	Printer	2	533.81	914 Main St	Credit Card	Pending	TRK13194259	3	NULL	Email	1067.62
8	ORD200031	2/4/2023	C26445	Chair	3	500.08	996 Main St	Debit Card	Returned	TRK31229981	6	NULL	Facebook	1500.24
9	ORD200032	2/5/2023	C12388	Tablet	5	536.72	830 Main St	Cash	Delivered	TRK99261395	10	WINTER15	Email	2683.6
10	ORD200034	2/7/2023	C67195	Chair	4	576.52	926 Main St	Credit Card	Pending	TRK55467915	5	SAVE10	Facebook	2306.08

Query 2: Sort Data Using ORDER BY

SQL Query:

SELECT Product, TotalPrice

FROM Sales

ORDER BY TotalPrice DESC

LIMIT 10;

Output:

	Product	TotalPrice
1	Tablet	3456.4
2	Monitor	3390.95
3	Laptop	3390.8
4	Chair	3384.9
5	Tablet	3370.2
6	Printer	3353.75
7	Laptop	3352.4
8	Printer	3334.0
9	Phone	3322.55
10	Laptop	3313.9

Query 3: Group Data Using GROUP BY

SQL Query:

SELECT Product, SUM(TotalPrice) AS TotalSales

FROM Sales

GROUP BY Product;

Output:

	Product	TotalSales
1	Chair	195620.11
2	Desk	167459.93
3	Laptop	192126.56
4	Monitor	175651.41
5	Phone	151722.39
6	Printer	195612.61
7	Tablet	186568.95

Query 4: COUNT

SQL Query:

SELECT COUNT(*) AS TotalOrders

FROM Sales;

Output:

	TotalOrders
1	1200

Query 5: SUM

SQL Query:

SELECT SUM(TotalPrice) AS TotalRevenue

FROM Sales;

Output:

	TotalRevenue
1	1264761.96

Query 6: AVG

SQL Query:

SELECT AVG(UnitPrice) AS AverageUnitPrice

FROM Sales;

Output:

	AverageUnitPrice
1	356.41275

Conclusion

This project demonstrates the use of SQL queries to retrieve, filter, sort, group, and analyze sales data using SELECT, WHERE, ORDER BY, GROUP BY, COUNT, SUM, and AVG.